

Trigonometric Functions

沈威宇

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1 Trigonometric Functions (三角函数)

I Trigonometric Ratios (三角比) or Trigonometric Functions

i Right triangle definitions for acute angles

Given a right triangle with an acute angle being theta:

- Sine (sin) (正弦):

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

- Cosine (cos) (餘弦):

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

- Tangent (tan) (正切):

$$\tan \theta = \frac{\text{hypotenuse}}{\text{adjacent}}$$

- Cotangent (cot) (餘切):

$$\cot \theta = \frac{1}{\tan \theta}$$

- Secant (sec) (正割):

$$\sec \theta = \frac{1}{\cos \theta}$$

- Cosecant (csc) (餘割):

$$\csc \theta = \frac{1}{\sin \theta}$$

ii Unit-circle definitions and properties

Let Γ be the ray obtained by rotating by an angle θ the positive half of the x -axis (counterclockwise rotation for $\theta > 0$ and clockwise rotation for $\theta < 0$). This ray intersects the unit circle with center $O = (0, 0)$ at the point $A = (x_A, y_A)$. Let the line extended to by Γ intersects $x = 1$ at $B = (1, y_B)$ and $y = 1$ at $C = (x_C, 1)$. The tangent line to the unit circle at A intersects the x - and y -axes at points $D = (x_D, 0)$ and $E = (0, y_E)$

Function	Symbol	Definition	Domain	Range	Period	Odd or even	Reflectional symmetric about	Rotationally symmetric of order 2 about
Sine	$\sin$	y_A	$\mathbb{R}$	$[-1, 1]$	2π	odd	$x \in \left\{x \mid \left(\frac{x}{\pi} - \frac{1}{2}\right) \in \mathbb{Z}\right\}$	
Cosine	$\cos$	x_A	$\mathbb{R}$	$[-1, 1]$	2π	even	$x \in \left\{x \mid \frac{x}{\pi} \in \mathbb{Z}\right\}$	
Tangent	$\tan$	y_B	$\mathbb{R} \setminus \left\{x \mid \left(\frac{x}{\pi} - \frac{1}{2}\right) \in \mathbb{Z}\right\}$	$\mathbb{R}$	π	odd		$(x, y) \in \left\{x \mid \left(\frac{x}{\pi} - \frac{1}{2}\right) \in \mathbb{Z}\right\} \times \{0\}$
Cotangent	$\cot$	x_C	$\mathbb{R} \setminus \left\{x \mid \frac{x}{\pi} \in \mathbb{Z}\right\}$	$\mathbb{R}$	π	odd		$(x, y) \in \left\{x \mid \frac{x}{\pi} \in \mathbb{Z}\right\} \times \{0\}$
Secant	$\sec$	x_D	$\mathbb{R} \setminus \left\{x \mid \left(\frac{x}{\pi} - \frac{1}{2}\right) \in \mathbb{Z}\right\}$	$(-\infty, -1] \cup [1, \infty)$	2π	even	$x \in \left\{x \mid \frac{x}{\pi} \in \mathbb{Z}\right\}$	$(x, y) \in \left\{x \mid \left(\frac{x}{\pi} - \frac{1}{2}\right) \in \mathbb{Z}\right\} \times \{0\}$
Cosecant	$\csc$	y_E	$\mathbb{R} \setminus \left\{x \mid \frac{x}{\pi} \in \mathbb{Z}\right\}$	$(-\infty, -1] \cup [1, \infty)$	2π	odd	$x \in \left\{x \mid \left(\frac{x}{\pi} - \frac{1}{2}\right) \in \mathbb{Z}\right\}$	$(x, y) \in \left\{x \mid \frac{x}{\pi} \in \mathbb{Z}\right\} \times \{0\}$

iii Exponential definitions

$$\begin{aligned}\sin x &= \frac{e^{ix} - e^{-ix}}{2i} \\ \cos x &= \frac{e^{ix} + e^{-ix}}{2} \\ \tan x &= -i \frac{e^{2ix} - 1}{e^{2ix} + 1}, \quad x \neq \frac{\pi}{2} + k\pi, k \in \mathbb{Z} \\ \cot x &= i \frac{e^{2ix} + 1}{e^{2ix} - 1}, \quad x \neq k\pi, k \in \mathbb{Z} \\ \sec x &= \frac{2e^{ix}}{e^{2ix} + 1}, \quad x \neq \pi + 2k\pi, k \in \mathbb{Z} \\ \csc x &= i \frac{2e^{ix}}{e^{2ix} - 1}, \quad x \neq 2k\pi, k \in \mathbb{Z}\end{aligned}$$

iv Power notation

$$\begin{aligned}\sin^n x &:= \begin{cases} (\sin x)^n, & n \geq 0 \\ \arcsin x, & n = -1 \end{cases} \\ \cos^n x &:= \begin{cases} (\cos x)^n, & n \geq 0 \\ \arccos x, & n = -1 \end{cases} \\ \tan^n x &:= \begin{cases} (\tan x)^n, & n \geq 0 \\ \arctan x, & n = -1 \end{cases} \\ \cot^n x &:= \begin{cases} (\cot x)^n, & n \geq 0 \\ \operatorname{arccot} x, & n = -1 \end{cases} \\ \sec^n x &:= \begin{cases} (\sec x)^n, & n \geq 0 \\ \operatorname{arcsec} x, & n = -1 \end{cases} \\ \csc^n x &:= \begin{cases} (\csc x)^n, & n \geq 0 \\ \operatorname{arccsc} x, & n = -1 \end{cases}\end{aligned}$$

v Trigonometric functions of important angles

Radian	Angle	sin	cos	tan
0	0°	0	1	0
$\frac{\pi}{2}$	90°	1	0	
π	180°	0	-1	0
$\frac{3\pi}{2}$	270°	-1	0	
$\frac{\pi}{4}$	45°	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
$\frac{3\pi}{4}$	135°	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	-1
$\frac{\pi}{6}$	30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
$\frac{\pi}{3}$	60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$\frac{2\pi}{3}$	120°	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$-\sqrt{3}$

Radian	Angle	sin	cos	tan
$\frac{5\pi}{6}$	150°	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{3}$
$\frac{\pi}{12}$	15°	$\frac{\sqrt{6}-\sqrt{2}}{4}$	$\frac{\sqrt{6}+\sqrt{2}}{4}$	$2-\sqrt{3}$
$\frac{5\pi}{12}$	75°	$\frac{\sqrt{6}+\sqrt{2}}{4}$	$\frac{\sqrt{6}-\sqrt{2}}{4}$	$2+\sqrt{3}$
$\frac{\pi}{10}$	18°	$\frac{\sqrt{5}-1}{4}$	$\frac{\sqrt{10+2\sqrt{5}}}{4}$	$\frac{\sqrt{5}-1}{\sqrt{10+2\sqrt{5}}}$
$\frac{2\pi}{10}$	36°	$\frac{\sqrt{10-2\sqrt{5}}}{4}$	$\frac{\sqrt{5}+1}{4}$	$\frac{\sqrt{10-2\sqrt{5}}}{\sqrt{5}+1}$
$\frac{3\pi}{10}$	54°	$\frac{\sqrt{5}+1}{4}$	$\frac{\sqrt{10-2\sqrt{5}}}{4}$	$\frac{\sqrt{5}+1}{\sqrt{10-2\sqrt{5}}}$
$\frac{4\pi}{10}$	72°	$\frac{\sqrt{10+2\sqrt{5}}}{4}$	$\frac{\sqrt{5}-1}{4}$	$\frac{\sqrt{10+2\sqrt{5}}}{\sqrt{5}-1}$
	37°	≈ 0.6018	≈ 0.7986	≈ 0.7536
	53°	≈ 0.7986	≈ 0.6018	≈ 1.3270

vi Sinusoid

A function $f(t) = A \cos(\omega t + \theta)$ for some real non-negative constants A, ω, θ is called a sinusoid.

vii Sinc function

The unnormalized sinc function is defined for $x \neq 0$ as

$$\text{sinc}(x) = \frac{\sin(x)}{x}.$$

The normalized sinc function is defined for $x \neq 0$ as

$$\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}.$$

II Phasor or complex amplitude

i Definition

The phasor or complex amplitude of a sinusoid $A \cos(\omega t + \theta)$ for some real non-negative constants A, ω, θ is a complex number

$$Ae^{i\theta}, \quad i = \sqrt{-1},$$

also denoted as $A\angle\theta$ or a two-dimensional vector

$$(A \cos \theta, A \sin \theta),$$

which satisfies

$$\Re(Ae^{i\theta}e^{i\omega t}) = A \cos(\omega t + \theta).$$

The $A\angle\theta$ notation is often called phasor and the $Ae^{i\theta}$ notation is often called complex number or complex amplitude.

Phasor denoting $A \cos(\omega t + \theta)$ is called rotating phasor. Phasor denoting $A \cos(\theta)$ is called non-rotating phasor.

ii Addition

Consider $X_1 \cos(\omega t + \varphi_1)$ and $X_2 \cos(\omega t + \varphi_2)$ and associated phasors $\underline{X}_1 = X_1 \angle \varphi_1$ and $\underline{X}_2 = X_2 \angle \varphi_2$.

$$X_1 \cos(\omega t + \varphi_1) + X_2 \cos(\omega t + \varphi_2) = X_m \cos(\omega t + \varphi),$$

where

$$X_m = |\underline{X}_1 + \underline{X}_2|$$

$$\varphi = \arg(\underline{X}_1 + \underline{X}_2)$$

iii Multiplication

$$(A \angle \theta)(B \angle \varphi) = (AB) \angle (\theta + \varphi)$$

iv Differentiation

The phasor of $\frac{d}{dt}(A \cos(\omega t + \theta))$ is $i\omega A e^{i\theta}$.

v Integration

The phasor of the sinusoid part of $\int A \cos(\omega \tau + \theta) d\tau$ is $-\frac{i}{\omega} A e^{i\theta}$.

vi Phasor diagram

A phasor as a two-dimensional vector on a plane is called phasor diagram.

III Inverse trigonometric functions (反三角函数)

i Definition

Function	Symbols	Definition	Domain	Range
Inverse sine (反正弦)	$y = \arcsin x = \sin^{-1}(x) = \text{asin}(x)$	$x = \sin y$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
Inverse cosine (反餘弦)	$y = \arccos x = \cos^{-1}(x) = \text{acos}(x)$	$x = \cos y$	$[-1, 1]$	$[0, \pi]$
Inverse tangent (反正切)	$y = \arctan x = \tan^{-1}(x) = \text{atan}(x)$	$x = \tan y$	$\mathbb{R}$	$(-\frac{\pi}{2}, \frac{\pi}{2})$
Inverse cotangent (反餘切)	$y = \text{arccot } x = \cot^{-1}(x) = \text{acot}(x)$	$x = \cot y$	$\mathbb{R}$	$(0, \pi)$
Inverse secant (反正割)	$y = \text{arcsec } x = \sec^{-1}(x) = \text{asec}(x)$	$x = \sec y$	$(-\infty, -1] \cup [1, +\infty)$	$[0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$
Inverse cosecant (反餘割)	$y = \text{arccsc } x = \csc^{-1}(x) = \text{acsc}(x)$	$x = \csc y$	$(-\infty, -1] \cup [1, +\infty)$	$[-\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2}]$

ii atan2 函數

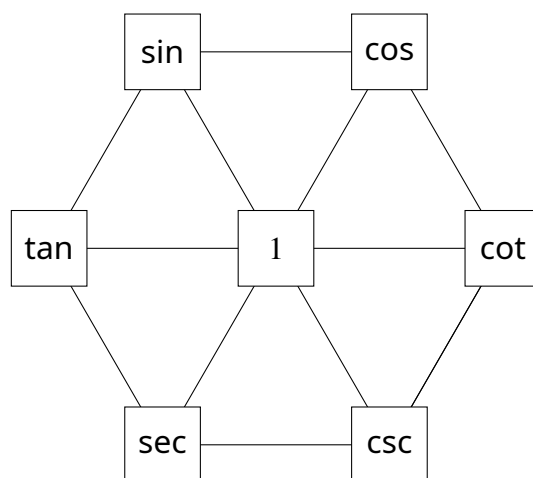
atan2 or $\text{arctan2}(y, x) : \mathbb{R}^2 \setminus \{(0, 0)\}$ 在 $x > 0$ 時返還 $\tan(\theta) = \frac{y}{x}$ 在 $(-\frac{\pi}{2}, \frac{\pi}{2})$ 中的解，在 $x < 0, y \geq 0$ 時返還 $\tan(\theta) = \frac{y}{x}$ 在 $(\frac{\pi}{2}, \pi)$ 中的解，在 $x < 0, y < 0$ 時返還 $\tan(\theta) = \frac{y}{x}$ 在 $(-\pi, -\frac{\pi}{2})$ 中的解，在 $x = 0, y \neq 0$ 時返還 $\frac{y}{|y|} \frac{\pi}{2}$ ，在 $x = y = 0$ 時未定義。

iii Trigonometric functions of inverse trigonometric functions

θ	$\sin \theta$	$\cos \theta$	$\tan \theta$	$\cot \theta$	$\sec \theta$	$\csc \theta$
$\arcsin(x)$	x	$\sqrt{1-x^2}$	$\frac{x}{\sqrt{1-x^2}}, x < 1$	$\frac{\sqrt{1-x^2}}{x}, x \neq 0$	$\frac{1}{\sqrt{1-x^2}}, x < 1$	$\frac{1}{x}, x \neq 0$
$\arccos(x)$	$\sqrt{1-x^2}$	x	$\frac{\sqrt{1-x^2}}{x}, x \neq 0$	$\frac{x}{\sqrt{1-x^2}}, x < 1$	$\frac{1}{x}, x \neq 0$	$\frac{1}{\sqrt{1-x^2}}, x < 1$
$\arctan(x)$	$\frac{x}{\sqrt{1+x^2}}$	$\frac{1}{\sqrt{1+x^2}}$	x	$\frac{1}{x}, x \neq 0$	$\sqrt{1+x^2}$	$\frac{\sqrt{1+x^2}}{x}, x \neq 0$
$\text{arccot}(x)$	$\frac{1}{\sqrt{1+x^2}}$	$\frac{x}{\sqrt{1+x^2}}$	$\frac{1}{x}, x \neq 0$	x	$\frac{\sqrt{1+x^2}}{x}, x \neq 0$	$\sqrt{1+x^2}$
$\text{arcsec}(x)$	$\frac{\sqrt{x^2-1}}{ x }$	$\frac{1}{x}$	$\sqrt{x^2-1} \text{sgn}(x)$	$\frac{\text{sgn}(x)}{\sqrt{x^2-1}}, x > 1$	x	$\frac{ x }{\sqrt{x^2-1}}$
$\text{arccsc}(x)$	$\frac{1}{x}$	$\frac{\sqrt{x^2-1}}{ x }$	$\frac{\text{sgn}(x)}{\sqrt{x^2-1}}, x > 1$	$\sqrt{x^2-1} \text{sgn}(x)$	$\frac{ x }{\sqrt{x^2-1}}$	x

IV Identities

i 三角函數基本關係



- 名稱：左側三者為正；右側三者為餘；上面二者為弦；中間二者為切；下面二者為割。
- 餘角關係：以鉛直軸為對稱軸，位於線對稱位置者，左 $(t) =$ 右 $(\frac{\pi}{2} - t)$ 。

$$\sin \theta = \cos \left(\frac{\pi}{2} - \theta \right)$$

$$\cos \theta = \sin \left(\frac{\pi}{2} - \theta \right)$$

$$\tan \theta = \cot \left(\frac{\pi}{2} - \theta \right)$$

$$\cot \theta = \tan \left(\frac{\pi}{2} - \theta \right)$$

$$\sec \theta = \csc \left(\frac{\pi}{2} - \theta \right)$$

$$\csc \theta = \sec \left(\frac{\pi}{2} - \theta \right)$$

- 倒數關係：三條通過中心點的線，其兩端者互為倒數，相乘為 1。

$$\sin \theta \csc \theta = 1$$

$$\cos \theta \sec \theta = 1$$

$$\tan \theta \cot \theta = 1$$

- 商數關係：六邊形周上，連續三個頂點形成的連線，其兩端者相乘等於中間者。

$$\tan \theta \cos \theta = \sin \theta$$

$$\sin \theta \cot \theta = \cos \theta$$

$$\sin \theta \sec \theta = \tan \theta$$

$$\cos \theta \csc \theta = \cot \theta$$

$$\tan \theta \csc \theta = \sec \theta$$

$$\cot \theta \sec \theta = \csc \theta$$

- 平方 (Pythagorean) 關係：圖中有三個倒正三角形，其上方兩頂點者之平方和等於在下方頂點者之平方。

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$\cot^2 \theta + 1 = \csc^2 \theta$$

ii 平移關係

$$\cos(x) = \sin\left(x + \frac{\pi}{2}\right) = -\sin\left(x - \frac{\pi}{2}\right)$$

$$\cot(x) = -\tan\left(x + \frac{\pi}{2}\right)$$

$$\csc(x) = \sec\left(x - \frac{\pi}{2}\right) = -\sec\left(x + \frac{\pi}{2}\right)$$

$$\arccos(x) = -\arcsin(x) + \frac{\pi}{2}$$

$$\operatorname{arccot}(x) = -\arctan(x) + \frac{\pi}{2}$$

$$\operatorname{arccsc}(x) = -\operatorname{arcsec}(x) + \frac{\pi}{2}$$

iii 奇變偶不變，正負看象限

今有函數 f ，已知其為 $\sin$ 、 $\cos$ 、 $\tan$ 、 $\sec$ 、 $\csc$ 、 $\cot$ 之一，且已知 $f(\theta)$ 。欲求 $f(\phi)$ ，其中 $\phi = \pm\theta \pm n\frac{\pi}{2}$ ， $n \in \mathbb{Z}$ 。

- 判斷方法：奇變偶不變，正負看象限。

- 上句：奇偶指 n 之奇偶，變指倒數，即：若 n 為奇數則令 $g(\theta) = \frac{1}{f(\theta)}$ ，否則令 $g(\theta) = f(\theta)$ ，則 $|f(\phi)| = |g(\theta)|$ 。

- 下句：象限指假設 $[r, \theta]$ 在第一象限時， $[r, \phi]$ 之象限。令該象限中任意角度為 ω 。令 $k = \frac{f(\phi)}{g(\theta)}$ 。則 $k = \frac{f(\omega)}{|f(\omega)|}$ ，即：

象限 f	一	二	三	四
sin	+	+	-	-
cos	+	-	-	+
tan	+	-	+	-
csc	+	+	-	-
sec	+	-	-	+
cot	+	-	+	-

iv 歐拉關係

$$\cos x + i \sin x = e^{ix}$$

$$\cos x - i \sin x = e^{-ix}$$

v 正切萬能公式

$$\sin \theta = \frac{2 \tan \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}}$$

$$\cos \theta = \frac{1 - \tan^2 \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}}$$

$$\tan \theta = \frac{2 \tan \frac{\theta}{2}}{1 - \tan^2 \frac{\theta}{2}}$$

vi 二倍角公式

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = 1 - 2 \sin^2 \theta$$

$$= 2 \cos^2 \theta - 1$$

$$= \cos^2 \theta - \sin^2 \theta$$

vii 半角公式或平方化倍角公式

$$\begin{aligned}\sin^2 \frac{\theta}{2} &= \frac{1 - \cos \theta}{2} \\ \cos^2 \frac{\theta}{2} &= \frac{1 + \cos \theta}{2} \\ \tan^2 \frac{\theta}{2} &= \frac{1 - \cos \theta}{1 + \cos \theta} \\ \tan \frac{\theta}{2} &= \frac{\sin \theta}{1 + \cos \theta} \\ &= \frac{1 - \cos \theta}{\sin \theta} \\ &= \frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta} \\ &= \csc \theta - \cot \theta\end{aligned}$$

viii 三倍角公式

$$\begin{aligned}\sin 3\theta &= 3 \sin \theta - 4 \sin^3 \theta \\ \cos 3\theta &= 4 \cos^3 \theta - 3 \cos \theta \\ \tan 3\theta &= \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}\end{aligned}$$

ix 和差角公式

$$\begin{aligned}\sin(\alpha + \beta) &= \sin \alpha \cos \beta + \cos \alpha \sin \beta \\ \sin(\alpha - \beta) &= \sin \alpha \cos \beta - \cos \alpha \sin \beta \\ \cos(\alpha + \beta) &= \cos \alpha \cos \beta - \sin \alpha \sin \beta \\ \cos(\alpha - \beta) &= \cos \alpha \cos \beta + \sin \alpha \sin \beta \\ \tan(\alpha + \beta) &= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \\ \tan(\alpha - \beta) &= \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} \\ \cot(\alpha + \beta) &= \frac{\cot \alpha \cot \beta - 1}{\cot \alpha + \cot \beta} \\ \cot(\alpha - \beta) &= \frac{\cot \alpha \cot \beta + 1}{\cot \beta - \cot \alpha} \\ \sec(\alpha + \beta) &= \frac{\sec \alpha \sec \beta}{1 - \tan \alpha \tan \beta} = \frac{\csc \alpha \csc \beta}{\cot \alpha \cot \beta - 1} \\ \sec(\alpha - \beta) &= \frac{\sec \alpha \sec \beta}{1 + \tan \alpha \tan \beta} = \frac{\csc \alpha \csc \beta}{\cot \alpha \cot \beta + 1} \\ \csc(\alpha + \beta) &= \frac{\csc \alpha \csc \beta}{\cot \alpha + \cot \beta} = \frac{\sec \alpha \sec \beta}{\tan \alpha + \tan \beta} \\ \csc(\alpha - \beta) &= \frac{\csc \alpha \csc \beta}{\cot \beta - \cot \alpha} = \frac{\sec \alpha \sec \beta}{\tan \alpha - \tan \beta}\end{aligned}$$

x 平方化正切平方公式

$$\sin^2 \theta = \frac{\tan^2 \theta}{1 + \tan^2 \theta}$$

$$\cos^2 \theta = \frac{1}{1 + \tan^2 \theta}$$

$$\cot^2 \theta = \frac{1}{\tan^2 \theta}$$

$$\sec^2 \theta = 1 + \tan^2 \theta$$

$$\csc^2 \theta = \frac{1 + \tan^2 \theta}{\tan^2 \theta}$$

xi 三角形內角正切公式

$$\tan \alpha + \tan \beta + \tan(\pi - \alpha - \beta) = \tan \alpha \cdot \tan \beta \cdot \tan(\pi - \alpha - \beta)$$

xii 正餘弦函數疊合

$$(a \sin \theta + b \cos \theta)^2 \leq a^2 + b^2, \quad a, b \in \mathbb{R}$$

$$\begin{aligned} a \sin x + b \cos x &= \sqrt{a^2 + b^2} \sin \left(x + \tan^{-1} \left(\frac{b}{a} \right) \right) \\ &= \sqrt{a^2 + b^2} \cos \left(x - \tan^{-1} \left(\frac{a}{b} \right) \right) \end{aligned}$$

xiii 和差化積公式

$$\sin \alpha + \sin \beta = 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$$

$$\sin \alpha - \sin \beta = 2 \cos \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$$

$$\cos \alpha + \cos \beta = 2 \cos \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$$

$$\cos \alpha - \cos \beta = -2 \sin \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$$

xiv 積化和差公式

$$2 \sin \alpha \cos \beta = \sin(\alpha + \beta) + \sin(\alpha - \beta)$$

$$2 \cos \alpha \sin \beta = \sin(\alpha + \beta) - \sin(\alpha - \beta)$$

$$2 \cos \alpha \cos \beta = \cos(\alpha + \beta) + \cos(\alpha - \beta)$$

$$2 \sin \alpha \sin \beta = -\cos(\alpha + \beta) + \cos(\alpha - \beta)$$

xv 連加公式

$$\sum_{k=1}^n \sin(k\theta) = \frac{\sin \left(\frac{n\theta}{2} \right) \sin \left(\frac{(n+1)\theta}{2} \right)}{\sin \left(\frac{\theta}{2} \right)}$$

$$\sum_{k=1}^n \cos(k\theta) = \frac{\sin \left(\frac{n\theta}{2} \right) \cdot \cos \left(\frac{(n+1)\theta}{2} \right)}{\sin \left(\frac{\theta}{2} \right)}$$

xvi 正餘切和等於正餘割積公式

$$\tan \theta + \cot \theta = \sec \theta \csc \theta$$

xvii 正餘弦四次方和公式

$$\sin^4 \theta + \cos^4 \theta = 1 - 2 \sin^2 \cos^2 \theta = 1 - \frac{1}{2} \sin^2(2\theta)$$

xviii 正餘弦四次方差公式

$$\sin^4 \theta - \cos^4 \theta = \sin^2 \theta - \cos^2 \theta = -\cos(2\theta)$$

xix 正餘弦六次方和公式

$$\sin^6 \theta + \cos^6 \theta = 1 - 3 \sin^2 \cos^2 \theta = 1 - \frac{3}{4} \sin^2(2\theta)$$

V 正弦連乘、餘切連加、餘割平方級數與餘切平方級數公式

$$\prod_{k=0}^{n-1} \sin \left(x + \frac{\pi k}{n} \right) = 2^{1-n} \sin(nx)$$

$$\sum_{k=0}^{n-1} \cot \left(x + \frac{\pi k}{n} \right) = n \cot(nx)$$

$$\sum_{k=0}^{n-1} \csc^2 \left(x + \frac{\pi k}{n} \right) = n^2 \csc^2(nx)$$

$$\sum_{k=1}^{n-1} \csc^2 \frac{\pi k}{n} = \frac{(n-1)(n+1)}{3}$$

$$\sum_{k=1}^{n-1} \cot^2 \frac{\pi k}{n} = \frac{(n-1)(n-2)}{3}$$

Proof.

$$\begin{aligned} & \prod_{k=0}^{n-1} \sin \left(x + \frac{\pi k}{n} \right) \\ &= \prod_{k=0}^{n-1} \frac{i}{2} \left(e^{-i \left(x + \frac{\pi k}{n} \right)} - e^{i \left(x + \frac{\pi k}{n} \right)} \right) \\ &= i^n 2^{-n} \prod_{k=0}^{n-1} e^{-i \left(x + \frac{\pi k}{n} \right)} \prod_{k=0}^{n-1} (1 - e^{2i \left(x + \frac{\pi k}{n} \right)}) \\ &= i^n 2^{-n} e^{-inx} e^{-i\pi \left(\frac{n-1}{2} \right)} \prod_{k=0}^{n-1} (1 - e^{2i \left(x + \frac{\pi k}{n} \right)}) \\ &= i^n 2^{-n} e^{-inx} i^{1-n} \prod_{k=0}^{n-1} (1 - e^{2i \left(x + \frac{\pi k}{n} \right)}) \end{aligned}$$

考慮：

$$f(t) = t^n - e^{2inx}$$

$f(t) = 0$ 的根為：

$$t = e^{2i\left(x + \frac{\pi k}{n}\right)}, \quad k \in \mathbb{N}_0 \wedge k < n$$

故：

$$f(t) = \prod_{k=0}^{n-1} (t - e^{2i\left(x + \frac{\pi k}{n}\right)})$$

$$\prod_{k=0}^{n-1} (1 - e^{2i\left(x + \frac{\pi k}{n}\right)}) = 1 - e^{2inx}$$

代回：

$$\begin{aligned} \prod_{k=0}^{n-1} \sin\left(x + \frac{\pi k}{n}\right) &= i^n 2^{-n} e^{-inx} i^{1-n} (1 - e^{2inx}) \\ &= 2^{-n} i (e^{-inx} - e^{inx}) \\ &= 2^{1-n} \sin(nx) \end{aligned}$$

$$\sum_{k=0}^{n-1} \ln \left| \sin\left(x + \frac{\pi k}{n}\right) \right| = (1-n) \ln(2) + \ln |\sin(nx)|$$

微分兩次：

$$\sum_{k=0}^{n-1} \cot\left(x + \frac{\pi k}{n}\right) = n \cot(nx)$$

$$\sum_{k=0}^{n-1} \csc^2\left(x + \frac{\pi k}{n}\right) = n^2 \csc^2(nx)$$

$$\sum_{k=1}^{n-1} \csc^2\left(x + \frac{\pi k}{n}\right) = n^2 \csc^2(nx) - \csc^2(x)$$

$$\sum_{k=1}^{n-1} \csc^2\left(\frac{\pi k}{n}\right) = \lim_{x \rightarrow 0} n^2 \csc^2(nx) - \csc^2(x) = \frac{(n-1)(n+1)}{3}$$

$$\cot^2(x) = \csc^2(x) - 1$$

$$\sum_{k=1}^{n-1} \cot^2 \frac{\pi k}{n} = \sum_{k=1}^{n-1} \csc^2 \frac{\pi k}{n} - n + 1 = \frac{n^2 - 3n + 2}{3}$$

□